

Control of chaos and its relevancy to spacecraft steering

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In 1990, a seminal work named controlling chaos showed that not only the chaotic evolution could be controlled, but also the complexity inherent in the chaotic dynamics could be exploited to provide a unique level of flexibility and efficiency in technological uses of this phenomenon. Control of chaos is also making substantial contribution in the field of astrodynamics, especially related to the exciting issue of low-energy transfer. The purpose of this work is to bring up the main ideas regarding the control of chaos and targeting, and to show how these techniques can be extended to Hamiltonian situations. We give realistic examples related to astrodynamics problems, in which these techniques are unique in terms of efficiency related to low-energy spacecraft transfer and in-orbit stabilization.

Keywords: chaos; nonlinear dynamics; celestial mechanics

1. Introduction

The dawn of understanding of the dynamics of chaos traces back to Henri Poincaré and his remarkable work in astrodynamics (Poincaré 1889). He introduced and analysed the presently famous circular restricted three-body problem. It gave him a glimpse of chaotic motion that appears on complicated and apparently unpredictable trajectories, which were close to periodic orbits, but spread fully in bounded regions of the phase space. Despite the important contributions made by mathematicians who have continued the studies of Poincaré, only in modern times, with the widespread use of computers with graphical capabilities, have the applications of chaotic dynamics become widespread. It is known now that the chaotic evolution is prevalent in nature and mediates numerous physical (Arecchi *et al.* 1982), biological (Schiff *et al.* 1994) and chemical phenomena (Hudson & Mankin 1981), among others.

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One contribution of 15 to a Theme Issue ‘Exploiting chaotic properties of dynamical systems for their control’.

In recent times, the legitimate desire to control natural phenomena, mediated by chaotic evolution, brought an exciting discussion as to whether it was possible to control its evolution. In 1990, a seminal work, named controlling chaos (Ott *et al.* 1990*a,b*), showed that not only the chaotic evolution could be controlled, but also the complexity inherent in the chaotic dynamics could be exploited to provide a unique level of flexibility and efficiency in the technological uses of these phenomena. The consequences of this work can be assessed by a number of subsequent works published on it. Control of chaos continues to be a very active area of research and is applied in all areas of science and technology. Novel approaches come up regularly, each one aiming to improve ongoing methods or extend their borders of usefulness.

Acknowledging the beginnings of chaos, the control of chaos is also making substantial contribution in the field of astrodynamics, particularly related to the important and timely issue of low-energy transfer (Boltt & Meiss 1995*a,b*; Macau 2000, 2003; Macau & Caldas 2002; Belbruno 2004). The goal behind this issue is to allow spacecraft transfer, but with a low consumption of propellant or energy. In this article, we show results on this issue that are directly based on the paradigm of controlling chaos, known as OGY strategy of chaos control.

Sensitive dependence on initial condition is the main characteristic of chaotic behaviour. This characteristic, known as the ‘butterfly effect’, can be used both to stabilize chaotic behaviour in periodic orbits and to direct trajectories rapidly to a desired state (Ott *et al.* 1990*a,b*; Shinbrot *et al.* 1990; Romeiras *et al.* 1992; Grebogi & Lai 1997). Ott *et al.* (1990*a,b*) showed that it is possible to stabilize a system’s chaotic evolution by exploring its own dynamical properties. Their approach uses only small perturbation applied to control parameters to stabilize chaos, keeping the parameters in the neighbourhood of their nominal values. The feasibility of the OGY method was experimentally demonstrated in several laboratory experiments, such as the ones by Ditto *et al.* (1990) and Garfinkel *et al.* (1992), and the OGY ideas were used in many other experiments (Azevedo & Rezende 1991; Bielawski *et al.* 1994; Meucci *et al.* 1994; Schiff *et al.* 1994). Today, these ideas are widely used in chaotic systems. Small perturbations can also be used to rapidly steer a chaotic system to a desired state. The inherent exponential sensitivity of chaotic time evolution to perturbations can be exploited to direct trajectories to some desired final state by using a carefully chosen sequence of small perturbations to some control parameter (Shinbrot *et al.* 1990). Here again, the perturbations can be so small that they do not significantly change the system dynamics, but enable the intrinsic system dynamics to drive the trajectory itself to the desired final state. This *targeting* process of guiding trajectories in chaotic systems also had its feasibility demonstrated experimentally (Shinbrot *et al.* 1992). Hereafter, we refer to this approach as chaotic targeting. A remarkable example of this process in astrodynamics was the guiding of the spacecraft ISEE-3/ICE (Van Roseninge *et al.* 1986) from a halo orbit to encounter with the Giacobini–Zinner comet, by using just a small amount of fuel for orbital correction.

The OGY method, which allows the stabilization of chaotic orbits, takes advantage of the fact that a chaotic set has a large number of unstable low-period periodic orbits embedded within it. Owing to transitivity, another characteristic of a chaotic system, chaotic trajectories visit the neighbourhood of these periodic

orbits. When the trajectory comes in the neighbourhood of a particular unstable periodic orbit (UPO), on which it is desired to stabilize the system, small perturbations to some accessible parameter are applied so that the state of the chaotic evolution is stabilized on the periodic orbit or, as the physicists say, the chaotic evolution is controlled. This method, which sparked a tremendous interest in the control of chaotic dynamical systems, is the outstanding example of how interesting it is to take advantage of the dynamical characteristics of the chaotic system in order to change its behaviour. Importantly, this process can be done from the data by using time-series analysis. The benefits of using the OGY method are multi-fold (Boccaletti *et al.* 2000). A chaotic system can be stabilized by using a very simple algorithm that requires a very small control effort.

In chaotic Hamiltonian systems, controlling and targeting are not easily accomplished. Besides the coexistence of interwoven chaotic and quasi-periodic regions, the phase space is divided into layered components, which are separated from each other by Cantori. Typically, a trajectory initialized in one layer of the chaotic region wanders in that layer for a long period of time before it crosses the Cantori to wander in the next layer. In this paper, we review how it is possible to overcome these difficulties in order to have an efficient chaos control and to derive targeting strategies that can be applied to Hamiltonian situations, exactly the situations that are faced in astrodynamical problems related to spacecraft transfer.

This paper is organized as follows. In §2, we review an extension of the OGY control of chaos method to Hamiltonian systems. In §3, we discuss a chaotic targeting method that can be applied to high-dimensional dissipative systems and also to Hamiltonian systems. In §§4 and 5, we apply these techniques to Hamiltonian systems that are considered as paradigm models for chaotic dynamics in astrodynamics: the restrict circular three-body problem and the Hill problem. In §7, we give the conclusions.

2. Stabilizing chaotic trajectories

One fundamental aspect of chaotic-invariant sets is that they are typically permeated by an infinite dense set of UPOs. The basic ideas of controlling chaos, using these UPOs, are as follows. First, one chooses a UPO embedded in the chaotic set according to some performance criteria. Second, one defines a small region around the desired periodic orbit. As a chaotic trajectory is transitive in its invariant set, starting from any initial condition, after some time it will come to the small region about the chosen UPO. When this occurs, small judiciously chosen parameter perturbations are applied to force and retain the trajectory evolving about the UPO. The OGY method, with small modifications, can be applied to hyperbolic Hamiltonian systems as follows (Lai *et al.* 1993*a,b*).

Let us consider a discrete time-dynamical system

$$\mathbf{X}_{i+1} = \mathbf{F}(\mathbf{X}_i, p), \quad (2.1)$$

where $\mathbf{X}_i \in \mathbf{X} \subset \mathbb{R}^2$, $p \in \mathbb{R}$ is an externally controllable parameter and $\mathbf{F}$ a smooth vector function in both variables. The parameter perturbation used to control the system is required to be small, i.e.

$$|p - p_0| < \delta, \quad (2.2)$$

where p_0 is the nominal parameter value and δ a small number defining the range of parameter variation. The goal is to programme the parameter p in such a way that a typical trajectory in the chaotic region is stabilized about the desirable UPO. The stabilization procedure starts to actuate only when the chaotic trajectory enters a small region around one of the periodic orbit points, whose size is proportional to δ . Once the particle is inside this small region, p is judiciously changed to keep the trajectory near the UPO.

Let us assume that the UPO of period n to be controlled is

$$\mathbf{X}_{01}(p) \rightarrow \mathbf{X}_{02}(p) \rightarrow \cdots \rightarrow \mathbf{X}_{0n}(p) \rightarrow \mathbf{X}_{0(n+1)}(p) = \mathbf{X}_{01}(p). \quad (2.3)$$

The linearized dynamics in the neighbourhood of the period- n orbit is

$$\mathbf{X}_{n+1} - \mathbf{X}_{0(n+1)}(p_n) = \mathbf{M}[\mathbf{X}_n - \mathbf{X}_{0n}(p_n)], \quad (2.4)$$

where $\mathbf{M}$ is the two-dimensional Jacobian matrix at the orbit point $\mathbf{X}_{0n}$ and $p_n = p_0 + (\Delta p)_n g_n$, where $(\Delta p)_n < \delta$ from equation (2.2). Let us observe that the parameter variation results in the following change in the periodic orbit points:

$$\mathbf{X}_{0n}(p_n) - \mathbf{X}_{0n}(p_0) \approx (\Delta p)_n \mathbf{g}_n, \quad (2.5)$$

where $\mathbf{g}_n = \partial \mathbf{X}_{0n}(p) / \partial p|_{p_0}$.

The Jacobian matrix $\mathbf{M}$ in equation (2.4) can be expressed in terms of its stable and unstable directions. It should be noted that even in the case in which $\mathbf{M}$ has complex-conjugate eigenvalues, the stable and unstable directions for $\mathbf{M}$ can be defined. To find the stable direction at a point $\mathbf{X}_0$, we first iterate the point N times in the forward direction under the map $\mathbf{F}$ and get the trajectory $\mathbf{X}_1 = \mathbf{F}(\mathbf{X}_0)$, $\mathbf{X}_2 = \mathbf{F}(\mathbf{X}_1) = \mathbf{F}^{[2]}(\mathbf{X}_0)$, ..., $\mathbf{X}_N = \mathbf{F}(\mathbf{X}_{N-1}) = \mathbf{F}^{[N]}(\mathbf{X}_0)$. Now, a circle of arbitrarily small radius ϵ is put at the point $\mathbf{X}_N$. If this circle is iterated backward once, it will become an ellipse at the point $\mathbf{X}_{N-1}$ with the major axis along the stable direction at the point $\mathbf{X}_{N-1}$. This ellipse is iterated backward, while at the same time its major axis is kept of the order of ϵ via certain normalization method. This procedure is repeated all the way back to the point $\mathbf{X}_0$, where the ellipse becomes very thin, with its major axis along the stable direction, provided N is large enough. This procedure is schematically shown in figure 1.

Similarly, as shown in figure 2, to find the unstable direction at point $\mathbf{X}_0$, first, the point is iterated backward under the inverse map N times to get a backward orbit $\mathbf{X}_{-j} = \mathbf{F}^{[-j]}(\mathbf{X}_0)$, ($j = 1, \dots, N$). At the point $\mathbf{X}_{-N}$, a circle of arbitrarily small radius ϵ is drawn about it. The previously discussed procedure is then applied, but iterating the point in the forward direction. At the end of this procedure, in $\mathbf{X}_0$, the ellipse becomes very thin, with its major axis along the unstable direction.

Let $\mathbf{e}_{s(n)}$ and $\mathbf{e}_{u(n)}$ be the stable and unstable directions at $\mathbf{X}_{0n}$, and let $\mathbf{f}_{s(n)}$ and $\mathbf{f}_{u(n)}$ be two vectors that satisfy $\mathbf{f}_{u(n)} \cdot \mathbf{e}_{u(n)} = \mathbf{f}_{s(n)} \cdot \mathbf{e}_{s(n)} = 1$ and $\mathbf{f}_{u(n)} \cdot \mathbf{e}_{s(n)} = \mathbf{f}_{s(n)} \cdot \mathbf{e}_{u(n)} = 0$. To control the orbit, it is required that the next iteration of a trajectory point, after falling into one of the small neighbourhoods around $\mathbf{X}_{0n}$, lies on the stable direction at $\mathbf{X}_{0(n+1)}(p_0)$, i.e.

$$[\mathbf{X}_{n+1} - \mathbf{X}_{0(n+1)}(p_0)] \cdot \mathbf{f}_{u(n+1)} = 0. \quad (2.6)$$

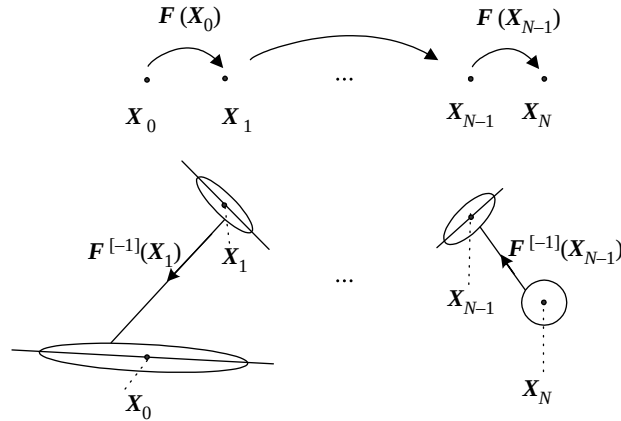


Figure 1. Geometric approach to find the stable direction associated with the point $\mathbf{X}_0$.

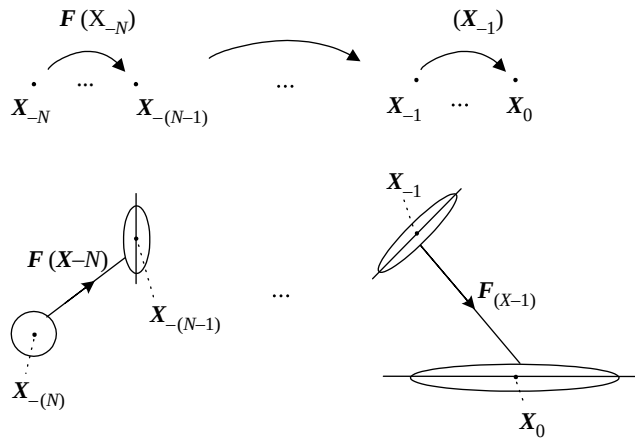


Figure 2. Geometric approach to find the unstable direction associated with the point $\mathbf{X}_0$.

Substituting equations (2.4) and (2.5) into equation (2.6), we obtain the following expression for the parameter perturbations:

$$(\Delta p)_n = \frac{\{M[\mathbf{X}_{n+1} - \mathbf{X}_{0(n+1)}(p_0)]\} \cdot \mathbf{f}_{u(n+1)}}{[(M\mathbf{g}_n) - \mathbf{g}_{n+1}] \cdot \mathbf{f}_{u(n+1)}}, \quad (2.7)$$

where M is evaluated at $\mathbf{X}_{0n}(p_0)$. This parameter perturbation is applied at each time-step of the trajectory, which is kept stabilized around the UPO.

3. Targeting: driving trajectories

The inherent exponential sensitivity of chaotic time evolution to perturbations is the hallmark of chaotic systems. This characteristic is responsible for the impossibility of making long-term predictions of the system evolution based on the finite precision measurement. However, despite the complexities of chaotic behaviour, the same main characteristic can be intelligently exploited to direct

the system to some desired state, using a carefully chosen sequence of small perturbations to some system parameter. This approach, which is of fundamental interest for the control system, is called *targeting* (Shinbrot *et al.* 1990).

The targeting idea came as a way to get around an excessive transient time associated with the use of the OGY method of chaos control to high-dimensional systems. As we saw in §2, the method relies on the topological transitivity of the system on the invariant set $\mathcal{A}$ to bring a chaotic orbit close enough to the neighbourhood of the periodic orbit on which we want to stabilize the system. This procedure works. Nevertheless, it presents a significant problem: the *transport time* can be excessively long. Besides, this time depends sensitively on the initial conditions and on the system's dimension. In dissipative chaotic systems, for randomly chosen initial conditions, the average transport time is typically ϵ^{-D} , where ϵ is the linear dimension of the neighbourhood about the periodic orbit, and D is the pointwise dimension at the periodic point (Kostelich *et al.* 1993). For low values of D , this time can be acceptably small; however, for systems of higher dimension, it may have a prohibitively larger value.

Let us consider a discrete time-dynamical system

$$\mathbf{X}_{i+1} = \mathbf{F}(\mathbf{X}_i, p), \quad (3.1)$$

where $\mathbf{X}_i \in \mathbb{R}^n$, $p \in \mathbb{R}$ is an externally controllable parameter that can be externally modified and $\mathbf{F}$ a smooth function in both variables. The nominal value of the parameter is $p = \bar{p}$, for which $\mathbf{F}$ is chaotic on a compact, invariant set $\mathcal{A} \subset \mathbb{R}^n$. Suppose we have two points $\mathbf{X}_s$ and $\mathbf{X}_t$ in $\mathcal{A}$. Let us consider a ball $B_\epsilon(\mathbf{X}_s)$ of radius ϵ around $\mathbf{X}_s$ and another ball $B_\epsilon(\mathbf{X}_t)$ of radius ϵ about $\mathbf{X}_t$. The *targeting* goal is to find a *constructive orbit* that goes from a point $\mathbf{p}\mathbf{x}_s \in B_\epsilon(\mathbf{X}_s)$ to $\mathbf{p}\mathbf{x}_t \in B_\epsilon(\mathbf{X}_t)$. Through this constructive orbit, the inherent exponential sensitivity of a chaotic time evolution to perturbations is intelligently exploited to direct trajectories to a desired state in the shortest possible time, by using a carefully chosen sequence of small perturbation to some control parameters. Furthermore, since these perturbations are sufficiently small, they do not significantly change the system's dynamics, but enable the intrinsic system dynamics to drive the trajectory to the desired state.

Our technique is subdivided into two sequential parts (Macau & Grebogi 1999). In the first part, we find the previously described points $\mathbf{p}\mathbf{x}_s$ and $\mathbf{p}\mathbf{x}_t$, so that there is an orbit (real) that goes from $\mathbf{p}\mathbf{x}_s$ to $\mathbf{p}\mathbf{x}_t$ (figure 3). In the second part, this orbit is used to build a constructive orbit (virtual) that allows the transfer from $\mathbf{p}\mathbf{x}_s$ to $\mathbf{p}\mathbf{x}_t$ using smaller number of elements, i.e. of real orbits. In this process, small perturbations to the control parameter are used to move among the real orbits. The effect of these perturbations is to change the system's evolution from one real orbit to another, resulting in a constructive orbit that allows the transfer from $\mathbf{p}\mathbf{x}_s$ to $\mathbf{p}\mathbf{x}_t$ at a faster time than the first part of our method. Thus, the overall effect of this procedure is to produce a sub-optimal solution that is obtained by the elimination of parts of the orbit, where *recurrences* (figure 4) occur with the use of small perturbations.

(a) Part I: finding a proper trajectory

The main idea of the first part of our technique is the following (Shinbrot *et al.* 1990, 1992): let us consider a line segment $\overline{a_1 b_1} \subset B_\epsilon(\mathbf{X}_s)$, so that $\mathbf{X}_s$ is its middle point (figure 3). To find $\mathbf{p}\mathbf{x}_s$, $\overline{a_1 b_1}$ is iterated in the forward direction, while the

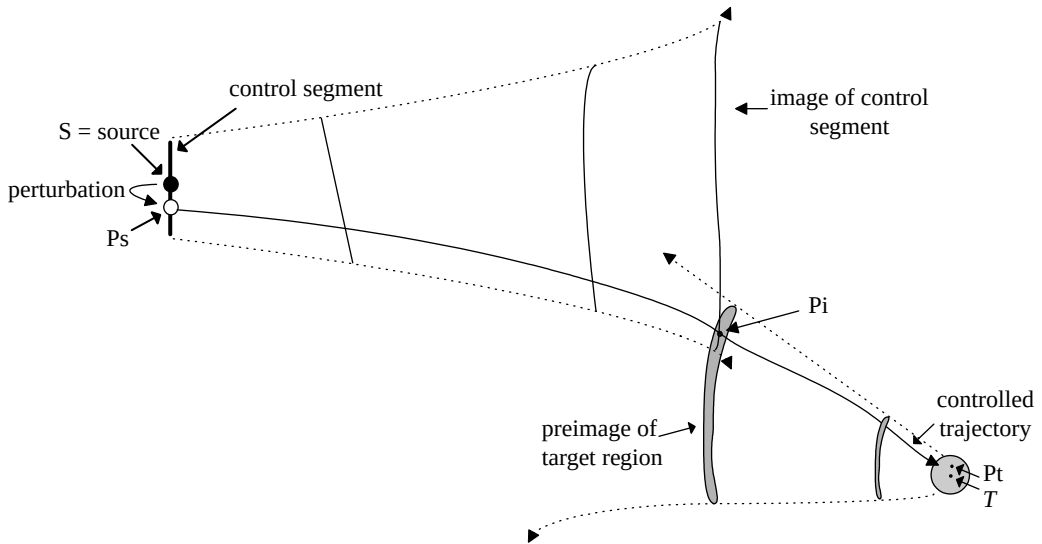


Figure 3. Schematic describing how to find a real, proper trajectory that goes from a nearby point of the source to the neighbourhood of the targeting point (part I of our targeting procedure).

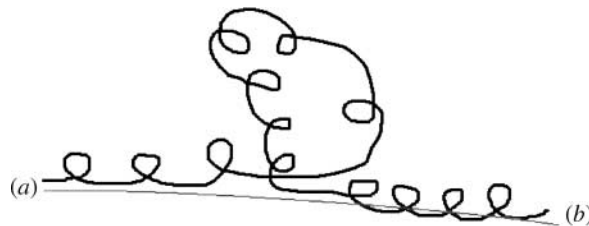


Figure 4. Schematic of the concept of recurrence and the *constructive orbit* or *pseudo-orbit* that is built by using small perturbations to allow the elimination of recurrent points.

region $B_\epsilon(x_t)$ is iterated in the backward direction, until the forward-iterated segment intersects the backward-iterated region at the point $\mathbf{p}_I$. It is important to say that it is again the transitivity of a chaotic system that ensures that $\mathbf{p}_I$ will be found. When the intersection is found, there is a trajectory that goes from $\mathbf{p}_{X_s} \in \overline{a_1 b_1}$ to $B_\epsilon(\mathbf{X}_t)$ through the intersection $\mathbf{p}_I$. It should be noted that $\mathbf{p}_{X_s}$ can be found by iterating $\mathbf{F}$ in the backward direction from $\mathbf{p}_I$. The point $\mathbf{p}_{X_s}$ is then used to determine the value $\tilde{p}$ of the parameter that must be applied to the system to bring it from $\mathbf{X}_s$ to $\mathbf{p}_{X_s}$ (figure 3). The following algorithm describes how this technique can be implemented:

Step 1. Define a direction Θ in space and use it to construct a line segment $\overline{a_1 b_1} \subset B_\epsilon(\mathbf{X}_s)$, so that $\mathbf{X}_s$ is its middle point. Call a_1 as c_0 and b_1 as c_{n_p} .

Step 2. Generate N random points $\{d_j\}_{j=1}^N$ inside $B_\epsilon(\mathbf{X}_t)$.

Step 3. Create a partition of n_p subsets in $\overline{c_0 c_{n_p}}$ using a sequence of $n_p - 1$ interior points $\{c_j\}_{j=1}^{n_p-1}$.

Step 4. Using $\{d_j\}_{j=1}^N$, construct a Delaunay triangulation (Watson 1981; Varosi *et al.* 1987) T , which has the sequence of cells $\{F_j\}_{j=1}^M$.

Step 5. Iterate in the forward direction $\{c_j\}_{j=1}^{n_p-1}$ and use linear interpolation to approximate the resultant curve delimited by each pair of iterated point.

Step 6. Iterate in the backward direction $\{d_j\}_{j=1}^N$ and use linear interpolation to approximate the iterated cells of Delaunay triangulation T .

Step 7. Continue the iteration described in steps 5 and 6 until finding the intersection $\mathbf{p}_I$ between $\mathbf{F}^l(c_k)\mathbf{F}^l(c_{k+1})$ and $\mathbf{F}^{-l}(F_h)$, where this is the backward-iterated cell found by linear interpolation of the backward iteration of the points that delineate the cell F_h .

Step 8. Consider the middle point c_{md} of the segment $\overline{c_k c_{k+1}}$. Identify whether the intersecting segment is $\overline{c_k c_{md}}$ or $\overline{c_{md} c_{k+1}}$. In the first case, assign the value of c_{md} to c_{k+1} , otherwise, assign the value of c_{md} to c_k . Applying a similar procedure, find a new cell F_h , which is smaller than the previous one, but still contains in its face p_{x_l} .

Step 9. If $d[c_k, c_{k+1}] > \delta_l$, where δ_l is a specified limit on the precision, then repeat step 8. Otherwise, $\mathbf{p}_{\mathbf{X}_s}$ is equal to c_k .

Step 10. Using $\mathbf{p}_{\mathbf{X}_s}$, determine the value of $\tilde{p}$ that drives the system from $\mathbf{X}_s$ to $\mathbf{p}_{\mathbf{X}_s}$. When the system gets to $\mathbf{p}_{\mathbf{X}_s}$, return the parameter to its nominal value, i.e. $\bar{p}$. From there, the system dynamics will conduct the system evolution to a point $\mathbf{p}_{\mathbf{X}_l} \in B_\epsilon(x_t)$ in $2 * l$ iterates.

With the use of this procedure, the average transport time to go from the source point to the target point typically scales logarithmically with the inverse of the size of the target region (Shinbrot *et al.* 1990), which contrasts with the exponential increases that take place if this algorithm is not used.

(b) Part II: finding a pseudo-orbit trajectory

Part I of our method produces an orbit that goes from $\mathbf{p}_{\mathbf{X}_s}$ to $\mathbf{p}_{\mathbf{X}_l}$. Let us represent this orbit by the following sequence of points $\{\mathbf{X}_i\}_{i=0}^N$ in $\mathbb{R}^n$, where $\mathbf{X}_0 = \mathbf{p}_{x_s}$ and $\mathbf{X}_N = \mathbf{p}_{x_l}$. As this orbit belongs to a chaotic trajectory in a compact invariant set A , it might have recurrent points (Boltt & Meiss 1995*a,b*). Here, we look for the *recurrent points* by using a sequential search (Macau & Grebogi 1999). If $\mathbf{X}_r$ is a recurrent point, it means that it belongs to a sequence of points $\{\dots, \mathbf{X}_r, \mathbf{X}_{r+1}, \dots, \mathbf{X}_{r+n}, \dots\}$ such that $d[\mathbf{X}_r, \mathbf{X}_{r+n}] < \delta$ makes up a kind of *loop*. If none of the points inside the loop is located in $B_\epsilon(\mathbf{X}_t)$, then the loop does not effectively conduct the trajectory to the targeting point. Thus, after being identified, our method replaces the loop by a smaller orbit that is backward asymptotic to $\mathbf{X}_r$ and forward asymptotic to $\mathbf{X}_{r+n}$ (Kostelich *et al.* 1993; Barreto *et al.* 1995). By creating patches similar to the recurrent points of the original orbit, we build a *constructive orbit* or a *pseudo-orbit* that allows the transfer from $\mathbf{p}_{\mathbf{X}_s}$ to $\mathbf{p}_{\mathbf{X}_l}$ with considerably less iterations than the original orbit. However, to accomplish it, perturbations must be introduced in order to switch the trajectory along the pseudo-orbits, as described in the following (Boltt & Meiss 1995*a,b*; Macau 1998).

In a hyperbolic situation, it is known that if the distance between $\mathbf{X}_r$ and $\mathbf{X}_{r+n}$ is sufficiently small, say less than δ_{lim} , then the unstable manifold of $\mathbf{X}_r$, $W_\epsilon^u(\mathbf{X}_r)$, and the stable manifold of $\mathbf{X}_{r+n}$, $W_\epsilon^s(\mathbf{X}_{r+n})$, intersect each other at a point $\mathbf{q}$. This fact can be exploited to accomplish our goal, if a proper perturbation is applied to the sequence of points of the original orbit that passes through $\mathbf{X}_r$. In fact, according to the theorem of Hirsh & Pugh (Arrowsmith & Place 1994),

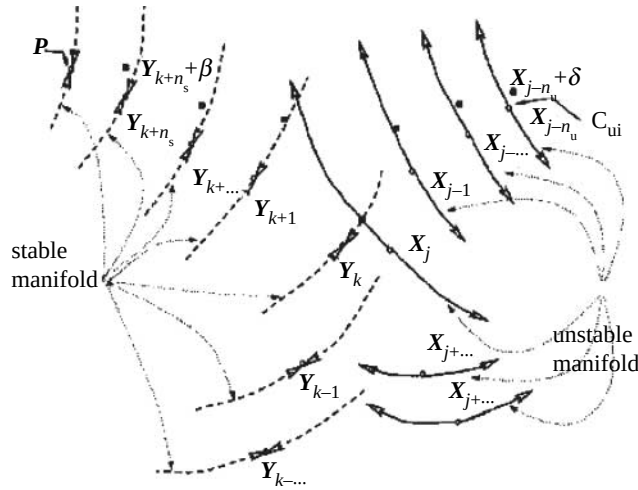


Figure 5. Schematic on how to find the proper perturbation to be used to cut recurrent points. With the applied β perturbation, the trajectory deviates from the original trajectory $\{Y_i\}$ and converges to the trajectory $\{X_i\}$.

$q \in W_\epsilon^u(X_r) \cap W_\epsilon^s(X_{r+n})$ implies that forward iterations of q converge to forward iterations of X_{r+n} , i.e. $\lim_{k \rightarrow \infty} d[F^k(q), F^k(X_{r+n})] = 0$, and backward iterations of q converge to backward iterations of X_r , i.e. $\lim_{k \rightarrow \infty} d[F^{-k}(q), F^{-k}(X_r)] = 0$. Thus, if we consider a point X_{r-m} that precedes X_r in the original trajectory, and a point X_{r+n+t} that succeeds X_{r+n} in the original trajectory, we have $d[F^t(q), F^t(X_{r+n})] = \epsilon_{r+n+t}$ and $d[F^{-m}(q), F^{-m}(X_r)] = \epsilon_{r-m}$. Furthermore, as $W_\epsilon^u(X_r)$ can be locally approximated by $E_{X_r}^u$, which is the unstable subspace of the tangent space at X_r , $W_\epsilon^s(X_{r+n})$ can be locally approximated by $E_{X_{r+n}}^s$, which is the stable subspace of the tangent space at X_{r+n} , and the approximation is continuously preserved over the iterations by the Jacobian of F , i.e. $DF(\cdot)$ calculated at the iteration point (Guckenheimer & Holmes 1983). It follows that ϵ_{r-m} is located in the direction of $E_{X_{r-m}}^u$ and ϵ_{r+n+t} is located in the direction of $E_{X_{r+n+t}}^s$. Thus, if the proper perturbation ϵ_{r-m} is applied in the direction of the $E_{X_{r-m}}^u$, it produces a perturbed orbit that passes through q and converges to the original trajectory after X_{r+n} . Consequently, this procedure generates the desired patch that avoids the recurrent loop of the original trajectory. In addition, this argument indicates that the perturbation ϵ_{r-m} can be calculated by solving the following equation (figure 5):

$$F^{m+t}(X_{r-m} + \epsilon_{r-m} E_{X_{r-m}}^u, \bar{p}) = X_{r+n+t} + \epsilon_{r+n+t} E_{X_{r+n+t}}^s, \quad (3.2)$$

which can be solved by using the Newton-secant method.

We should emphasize that the values of m and t in equation (3.2) can be adequately adjusted for each system by an empirical procedure. Hirsh & Pugh's theorem also provides us with a proper way to use the approximation of the tangent subspace $E_{X_i}^s$ and $E_{X_i}^u$ at a point. According to this theorem, if we consider an orbit $\{X_k\}_{k=1}^n$, which contains X_i , any variation near X_{i-m} will expand along the unstable manifold of X_i , if m is chosen large enough. A similar statement can be made regarding the stable manifold of X_i for variations near X_{i+m} , iterated in the backward direction.

This procedure can be used in an attempt to eliminate the recurrence in the original path from $\mathbf{px}_s$ to $\mathbf{px}_t$ that are less than δ_{lim} . Higher priority in the elimination should be assigned to the longest loops. A patch is accepted as usable, if the perturbation ϵ_{r-m} to be applied, in order to implement it, is less than a pre-assigned limit value ϵ_{lim} . Our method spawns a sequence of perturbations $\{\epsilon_i\}_{i=1}^K$ and directions $\{\mathbf{E}_i\}_{i=1}^K$ to be, respectively, applied to a sequence of points $\{\mathbf{X}_{n_i}\}_{i=1}^K$ of the original trajectory. To apply each perturbation, it is necessary to calculate the value $\widetilde{\mathbf{px}_{n_i}}$ of the parameter to be used in $\mathbf{X}_{n_i}$, to change the system state from $\mathbf{X}_{n_i}$ to $\mathbf{X}'_{n_i} = \mathbf{X}_{n_i} + \epsilon_i \mathbf{E}_{x_i}$. The overall result of our method is a sub-optimal constructive trajectory or a sub-optimal pseudo-orbit that allows the transfer between $\mathbf{px}_s$ and $\mathbf{px}_t$.

The previous arguments can be consolidated in the following algorithm:

Step 1. Starting from the original transfer trajectory from $\mathbf{px}_s$ to $\mathbf{px}_t$, find all the recurrent points whose distance from the trajectory is less than δ_{lim} . Sort them out by the size of the loop in decreasing order.

Step 2. Take from the list its first point and find a patch for the loop using equation (3.2). If the resulting perturbation is less than ϵ_{lim} , accept the patch. Put in the solution list the points in which the perturbation should be applied, together with the perturbations and the direction values.

Step 3. Take the next point in the list that is located after the previously found patch.

Step 4. Find a patch for the loop using equation (3.2). If the resulting perturbation is less than ϵ_{lim} , accept the patch. Put in the solution list the points in which the perturbation should be applied, together with the perturbations and the direction values.

Step 5. Go back to step 3 until all the points of the list have been considered.

Step 6. Use the solution list and the original trajectory to compute the pseudo-orbit that allows the sub-optimal transfer between $\mathbf{px}_s$ and $\mathbf{px}_t$.

(c) The targeting algorithm

The algorithm that results from the combination of parts I and II can be applied to general situations (Macau & Grebogi 1999). In fact, separately, both parts have been successfully applied to numerical and laboratory experiments on mechanical (Shinbrot *et al.* 1994; Kostelich *et al.* 1993) and on situations involving spacecraft guidance (Boltt & Meiss 1995b). Furthermore, with delay-coordinate embedding, the algorithm is applicable to experimental situations, in which no *a priori* analytical knowledge of the system dynamics is available (Shinbrot *et al.* 1992).

The power of our method is due to the sequential combination of both parts. However, we must stress the fact that the second part has the objective of reducing long trajectories that present recurrence. We can have situations where the algorithm does not succeed because there are no recurrences in the trajectory for the specified limit values of the perturbation and the proximity between the recurrent points. In other situations, the trajectory found by the first part of the algorithm is short enough to satisfy our goals. Nevertheless, in general, the second part of the algorithm is always necessary to be used for high-dimensional or Hamiltonian systems, but not for low-dimensional systems.

4. Targeting in the restrict three-body problem

We now consider targeting in the planar and circular restricted three-body problem (Roy 1988) to model the dynamics of a spacecraft moving in the Earth–Moon system. This problem is the special case of the full three-body problem, in which one of the masses is taken to be infinitesimal, and so has no influence on the two primaries, which are in circular orbits. We use the standard coordinates for the restrict three-body problem, and normalize the sum of the masses to one, i.e. $m_e + m_m = 1$, and Newton's gravity constant to one. The positions of the primaries in the rotating frame are fixed at $x_e = -m_m$ and $x_m = m_e$. In this frame, the equations of motion for the spacecraft are the following:

$$\dot{x} = u, \quad \dot{y} = v, \quad (4.1)$$

$$\dot{u} = x + 2v - m_e \frac{x - x_e}{r_e^3} - m_m \frac{x - x_m}{r_m^3}, \quad (4.2)$$

$$\dot{v} = y - 2u - (x^2 + y^2) - 2\left(\frac{m_e}{r_e} + \frac{m_m}{r_m}\right), \quad (4.3)$$

where x and y are the position coordinates in the rotating frames; u and v the corresponding momenta; and r_m and r_e denote the distances of the spacecraft from the fixed position of m_m and m_e , respectively. In the case of the Earth–Moon system, the mass ratio is $m_m/m_e = 0.0123$ (Roy 1988). The Jacobi integral

$$J = u^2 + v^2 - (x^2 + y^2) - 2\left(\frac{m_e}{r_e} + \frac{m_m}{r_m}\right), \quad (4.4)$$

is a constant of the motion, confining the Hamiltonian flow to a three-dimensional sub-manifold, $J = \text{constant}$. The parameters time and velocity are measured in units of 104 h and 1024 m s^{-1} , respectively.

We introduce (Macau 2000) the Poincaré section $y = 0$, mapping the (x, u) -plane to itself, whenever the trajectory traverses the Poincaré section with $v > 0$. Thus, the Poincaré return map defined for fixed J is a two-dimensional area-preserving map.

The classical way to transfer a spacecraft from a circular parking orbit, near the Earth, to a small-radius circular lunar orbit is by using the *Hohmann transfer* (Bate *et al.* 1971). This transfer typically takes only a few days, and requires two large rocket thrusts, one to accelerate the spacecraft to leave the Earth, and another to decelerate it relative to the Moon to achieve lunar capture. In this scenario, the use of a 'chaotic transfer', as we will see, reduces the overall fuel consumption and, so, decreases the transport cost significantly. This problem has been addressed previously by Boltt & Meiss (1995b). They started from a circular parking orbit around the Earth. A first rocket thrust is used to put the spacecraft into a chaotic phase space region that extends to the vicinity of the Moon. The spacecraft follows a trajectory that would eventually approach an orbit forward asymptotic to a Moon-orbiting invariant torus. However, this transfer would be prohibitively long. Their strategy removes recurrent loops from such a long trajectory by using small control thrusts. Boltt & Meiss substantially reduced the transfer time to 2.05 years by using this method, while saving 40% of the thrust compared to the Hohmann transfer from the same parking orbit.

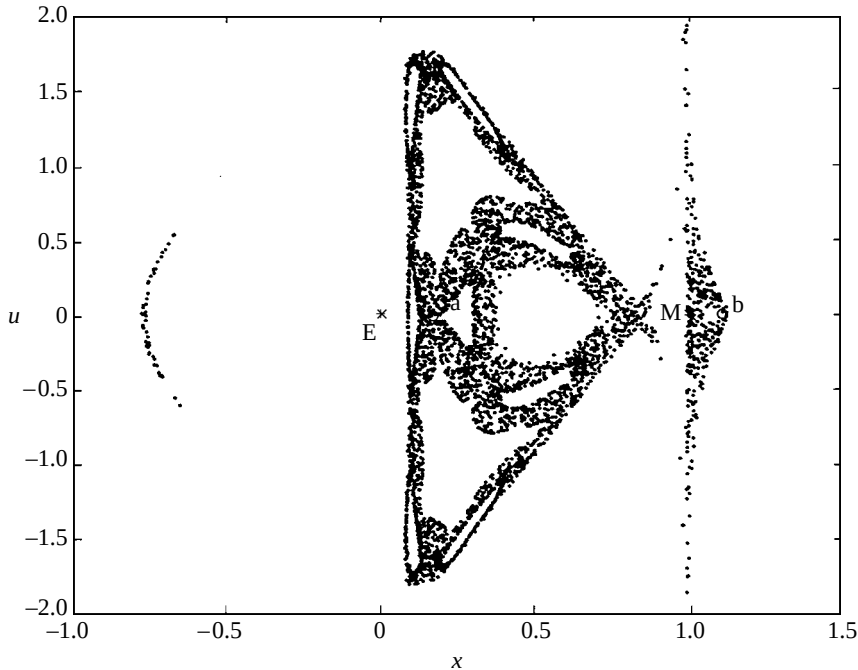


Figure 6. Poincaré surface of the planar-restricted three-body problem at $J = -3.17948$. The source and the target points are denoted by a and b , respectively.

Let us now apply our method. We assume that the spacecraft is in a circular parking orbit around the Earth of radius 59.699 km. From this orbit, it is necessary to apply an impulsive thrust to move the spacecraft to the chaotic region. To accomplish this, we give to it a change in velocity of $\Delta V_{\text{control}} = 744.4 \text{ m s}^{-1}$. After this manoeuvre, the spacecraft is in the point a in the Poincaré section (figure 6). From this point, we have a very long chaotic trajectory that eventually comes close to the Moon. This trajectory is depicted in figure 6. Part I of our method can now be applied, resulting in the point $\hat{a}$ in the Poincaré section (figure 6). This point belongs to a real trajectory that comes close to the Poincaré section of the stable orbit around the Moon that we wish to target. This stable orbit is represented by the point b , as shown in figure 6. A velocity of $\Delta V_{\text{control}} = 2.053 \times 10^{-2} = 21.02 \text{ m s}^{-1}$ is required to move the spacecraft from a to $\hat{a}$ (Macau 2000). **At the point where this transfer trajectory to the Moon comes closest to the point b in the Poincaré section, we apply** the procedure described in part II of §3 to get the stabilizing thrust required to move the spacecraft to a stable orbit located at b . Actually, this orbit was the only recurrent trajectory point found in part I. This last thrust is $\Delta V_{\text{stab.}} = 2.32 \times 10^{-3} = 2.38 \text{ m s}^{-1}$. The corresponding transfer orbit is shown in figure 7. It takes about 284 days for the transfer (Macau 2000).

The overall thrust required to move the spacecraft from the parking orbit around the Earth to the stable orbit around the Moon is $\Delta V = 767.8 \text{ m s}^{-1}$, which is above 749.6 m s^{-1} required by the transfer in Boltt & Meiss (1995*a, b*). However, we also got a considerable shorter transfer time, but it was necessary to apply two impulsive thrusts.

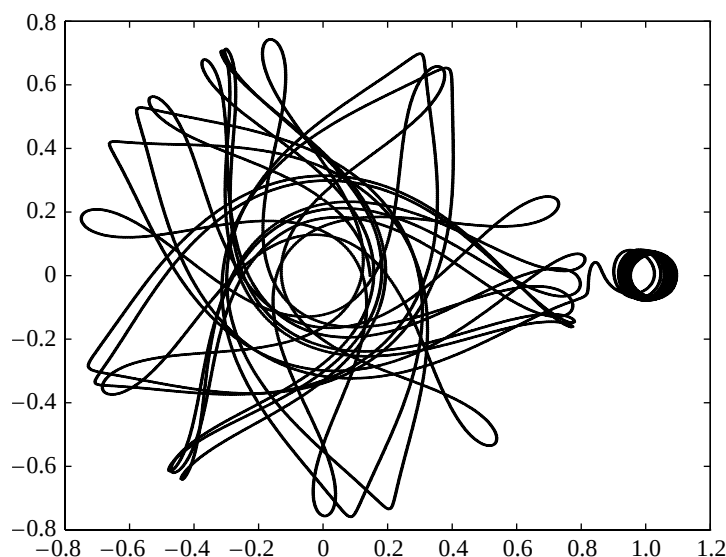


Figure 7. Using our targeting method, the spacecraft is guided to a stable orbit around the Moon, which takes about 284 days for the transfer.

5. The Hill problem

We now consider two light bodies M_2 and M_3 describing initially coplanar and circular orbits, with slightly different radii, around a heavy central body M_1 (Macau 2003; figure 8). If the two light bodies are far apart, their mutual attraction can be neglected, and the problem reduces, in a reasonable approximation, to a superposition of two independent two-body problems. However, if the distance between them becomes sufficiently small, a situation called an *encounter*, their mutual attraction is no longer negligible and cannot be ignored. This is known as *Hill's problem* (Hill 1878).

Hill's problem can be considered as the simplest non-integrable case of the N -body problem. The equations are very simple and contain no parameter. However, many problems in celestial mechanics can be adequately approximated by Hill's equations. Examples are: the Sun–Earth–Moon problem (Hill 1878), which was the motivation of Hill's original work; the interaction between particles in planetary rings (Goldreich & Tremaine 1979; Hénon 1981); the motion of co-orbital satellites (Dermontt & Murray 1981); the accretion of particles by a proto-planet (Schofield 1981); and the distribution of particles around the Earth (Dole 1962).

We consider Hill's problem for the case where M_2 is a natural satellite of M_1 , while M_3 is an artificial satellite and our goal is to stabilize the latter in a convenient orbit about M_2 (Macau 2003). The difference between the radii of their initial circular orbits is called the *impact parameter*, and represented by the parameter h . In the initial situation, M_2 , the inner body has a slightly larger angular velocity than M_3 , which means that after some time eventually an encounter occurs. In Macau & Caldas (2002), we have analysed the dynamics of this problem, showing that there are regions of the impact parameter where a

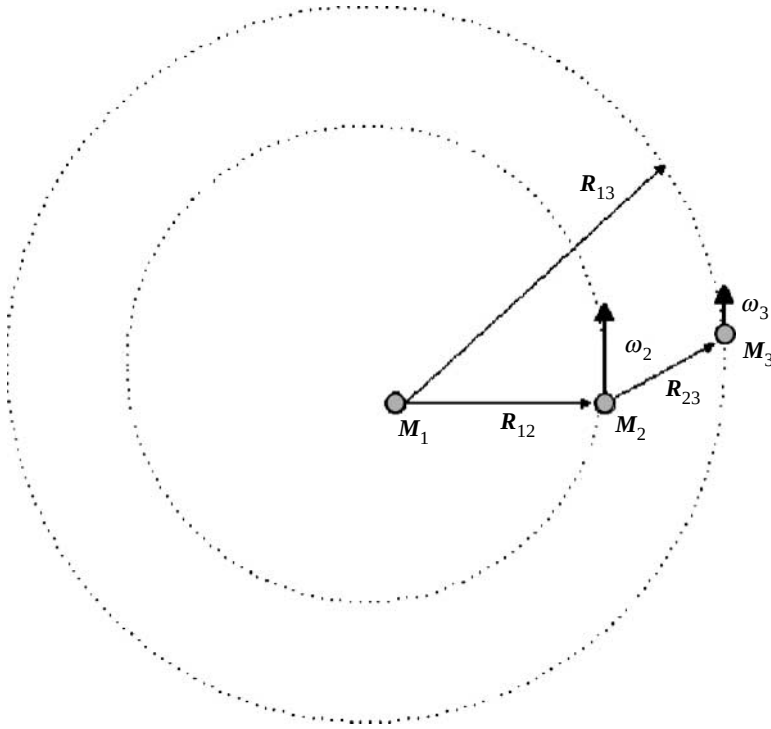


Figure 8. Two light bodies describing initially coplanar and circular orbits, with slightly different radii, around a heavy central body. In the initial situation M_2 , the inner body has a slightly larger angular velocity than M_3 , meaning that after some time, they eventually interact gravitationally with each other.

chaotic scattering appears. In these regions, it is possible to stabilize the chaotic evolution of the satellites around one of the UPOs of the chaotic evolution using the previously discussed method.

Let us assume that the mass of either satellite is small when compared with the mass of the planet, i.e.

$$m_1 \gg m_2 \quad \text{and} \quad m_1 \gg m_3, \quad (5.1)$$

where m_i is the mass of body M_i . The ratio of the two masses m_2 and m_3 can be arbitrary, but fixed. Thus, we define

$$m = m_1 + m_2 + m_3, \quad (5.2)$$

and

$$\mu = \frac{m_2 + m_3}{m_1}. \quad (5.3)$$

We consider that the distance between the two satellites is small when compared with their distance to the planet. Thus, the two satellites can be approximately viewed as a single body in orbit around the planet. We call this orbit the *mean orbit* (Henon & Petit 1986; Petit & Henon 1986), and assume it to be a circular orbit with radius a_0 . The angular velocity of m_2 and m_3 in this mean orbit is

$$\omega = \sqrt{Gma_0^{-3}}. \quad (5.4)$$

In Henon & Petit (1986) and Petit & Henon (1986), it is shown that when the satellites are moving in the neighbourhood of each other and the movement is described using synodic coordinates, a sequence of transformations can be applied. The results are Hill's equations for the relative motion of the satellites, described in ξ and η coordinates:

$$\dot{\xi} = u, \quad \dot{\eta} = v, \quad \dot{u} = 2v + 3\xi - \frac{\xi}{\rho^3}, \quad \dot{v} = -2u - \frac{\eta}{\rho^3}, \quad (5.5)$$

where

$$\rho = \sqrt{\xi^2 + \eta^2}. \quad (5.6)$$

Hill's equations admit the integral

$$\Gamma = 3\xi^2 + \frac{2}{\rho} - u^2 - v^2, \quad (5.7)$$

which is called the *Jacobi integral* by analogy with the restricted problem. The Jacobi integral can be written in terms of the initial conditions (impact parameter)

$$\Gamma = \frac{3}{4}h^2. \quad (5.8)$$

We can show that for this problem, in some regions of the parameter h , there is a chaotic-invariant set related to the phenomenon known as **chaotic scattering**. **Chaotic scattering** is characterized by 'sensitive dependence' of output variables that characterize the particle trajectory after the scattering to small changes in an input variable that characterizes the trajectory before scattering (Bleher *et al.* 1990). The dynamics of a chaotic scattering can be explained by the existence of a chaotic-invariant saddle (Hsu *et al.* 1988), which is the intersection of two Cantor sets of roughly parallel surfaces. Embedded in this chaotic-invariant set, there is an infinite, numerable and dense set of UPOs. Using the control of chaos methods, these orbits can be used to control the orbit of one satellite around the other, as shown in §6.

6. Controlling a satellite encounter

To apply our method, we introduce the Poincaré section $\eta = 0$, mapping the (ξ, u) -plane to itself, whenever the trajectory traverses the Poincaré section with $v < 0$. Thus, the Poincaré return map for fixed Γ is a two-dimensional map (figure 9).

We use the Schmelcher & Diakonou (1998) method, in association with the Newton–Raphson method, to find periodic orbits for the values of the impact parameter for which the chaotic scattering is present (Macau 2003). The first method was applied to each point of a set of starting points defined by a uniform grid of initial conditions over the state space. For each point of this set, the method converges to a solution. From the application of this method, we get the solution points of the desired periodic orbits and also points of quasi-periodic orbits. All these points are then used as starting points for the Newton–Raphson method.

Trajectories very close to the UPO follow it for a while, after which they escape from it by following the unstable manifold of that periodic orbit. Thus, in our problem situation, one satellite can be kept in orbit around the other following the UPO just if a control algorithm for stabilization was applied. Since we are considering a situation for which the system presents a chaotic scattering,

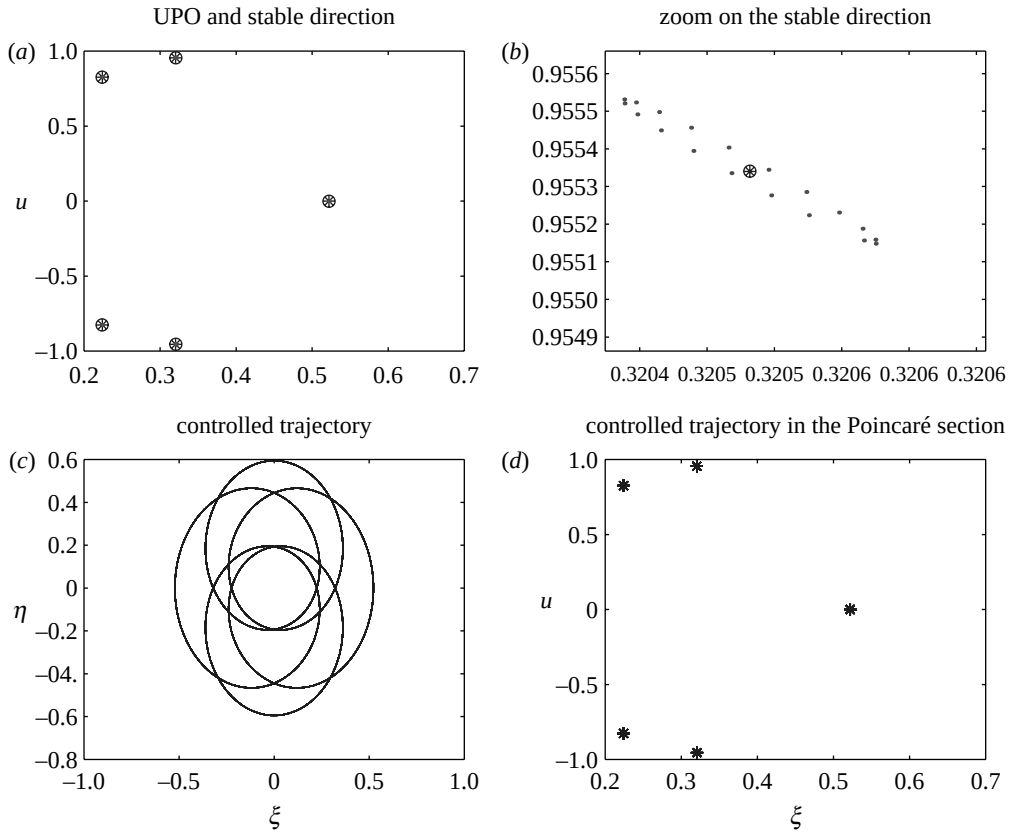


Figure 9. (a) The unstable periodic orbit embedded in the chaotic-invariant set that was chosen as a desired orbit to be stabilized; (b) the stable direction found by using geometric approach (compare with figure 1); (c) the controlled orbit; and (d) the controlled orbit in the Poincaré section.

which means that the UPO is embedded in a non-attracting chaotic-invariant set, we can apply the discussed control of chaos technique (Macau 2003). The result of applying this stabilization strategy is shown in figure 9.

7. Conclusion

The purpose of this paper was to lay out the main ideas regarding the control of chaos and targeting, to show how these techniques can be extended to Hamiltonian situations, and to present applied examples related to astrodynamics problems, in which these techniques are unique in terms of efficiency related to low-energy spacecraft transfer and in-orbit stabilization. Sensitive dependence on initial conditions is the main characteristic of chaotic behaviour. The efficiency regarding spacecraft transfer at the expense of small amount of fuel consumption comes from using just small perturbations or very low intense thrust to accomplish spacecraft manoeuvre whenever chaos is present. As the

lifetime of an automatic space exploration mission is mainly defined by the amount of fuel it can carry, the direct benefits of chaotic transfer in astrodynamics are twofold. It can allow a spacecraft to go farther in exploring the limits of our Solar System, or it can save the amount of fuel, meaning a larger amount of scientific instruments in the payload, since the liftoff capacity of a launch vehicle is limited. Furthermore, these techniques can also be used to rescue spacecrafts that end up in a wrong trajectory accidentally due to defects in the launch phase. Actually, two space missions were successfully rescued by using chaos-based transfer, as was the case with the Japanese spacecraft Hiten, in 1991 (Belbruno 1994), and the American communication satellite HGS-1, in 1998.

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